Blueprint: The Case I/II engine and Washington 9.5 descent (flt-vandiver)

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Abstract

This blueprint maps `flt-vandiver`, the engine of the development: it reduces `FermatLastTheoremFor` p to two finite certificates, discharging Case II by a machine-checked instance of Washington's Theorem 9.5 small-witness descent and Case I by a Bernoulli or Sophie Germain criterion. Regularity is never used; the weaker input $p \nmid h^+$ is supplied by a Q_i cyclotomic-unit certificate.

Two inputs are consumed from sibling libraries and appear here as external leaves: the cyclotomic-unit index $[E : C] = h^+$ (from `flt-cyclotomic-nt`) and Hilbert's Theorem 94 with Kummer-extension unramifiedness (from `flt-regular`).

Nodes are linked by `\lean` to their declarations (namespaces `FltVandiver`, `FltVandiver.Descent95`); this is a milestone-level map of the engine, centered on the Case II descent.

0.1 The Q_i certificate and the Vandiver bridge

Definition 0.1.1 (Q_i / Vandiver certificate). *The Boolean certificate* `vandiverCert` $p\,\ell\,t$ *asserting that the* Q_i *cyclotomic-unit test fires at every even index for the auxiliary pair* (ℓ, t)*.*

Definition 0.1.2 (Fast sliced certificate). *A pure-*`Nat` *reformulation* `vandiverCertFast` *of the certificate, with exponents reduced modulo* $\ell - 1$*, that* `native_decide` *evaluates for every prime below* 1000*.*

Definition 0.1.3 (Machine-speed certificate). *The evaluator* `fast2Cert` *of the all-even test by discrete-logarithm sums:* $Q_i^k = h^{S_i - Nd_i}$ *with* $h = t^k$ *of order* p*, so an index costs* $p/2$ *multiplications modulo* p *once* $O(p)$ *tables are built. Every table fact it uses is re-verified inside the Boolean. Precompiled; it carries the two Wolstenholme primes (*16843 *in seconds,* 2124679 *in* 31 *core-hours).*

Theorem 0.1.4 (Machine-speed certificate ⇒ certificate). *If* `fast2Cert` $p\,\ell\,t\,idx$ *holds then so does* `vandiverCert` $p\,\ell\,t\,idx$*: the two evaluators agree on what they certify, on the three standard axioms.*

Theorem 0.1.5 (Base independence of the Q_i test). *For two valid bases* t, t' *and an even index* $i < p$*,* $Q_i(t)^k = 1$ *if and only if* $Q_i(t')^k = 1$*: the verdict of the test depends on* (p, ℓ) *alone, so a passing certificate at every even index passes for every valid base (*`vandiverCert_evenIndices_of_base`*). A witness is the prime* ℓ*.*

Definition 0.1.6 (Unit index $[E : C] = h^+$ (external)). *Washington's Theorem 8.2,* `cyclotomic_unit_index` *in* `flt-cyclotomic-nt`*: the index of the cyclotomic units equals* h^+*. It turns the* Q_i *test into a statement about* $p \nmid h^+$*.*

Theorem 0.1.7 (Q_i certificate ⇒ Vandiver). *If* `vandiverCert` $p\,\ell\,t$ *holds then* $p \nmid h^+$*: the* Q_i *test firing at every even index rules out each eigen cyclotomic unit being a* p*-th power.*

Definition 0.1.8 (Small-witness certificate). `SmallWitnessCert` p *packages the pair* (ℓ, t) *with* $\ell < p^2 - p$ *and the* Q_i *certificate — the single finite input that powers both the Vandiver conclusion and the Case II descent.*

0.2 Case II: the Washington 9.5 small-witness descent

Definition 0.2.1 (Hilbert 94 / Kummer unramifiedness (external)). *Hilbert's Theorem 94 and the unramifiedness of the relevant Kummer extension, consumed from* `flt-regular`*. These seed the* $\ell \mid \xi$ *invariant that the descent tracks.*

Theorem 0.2.2 (Kummer extension unramified). *The Kummer extension attached to a putative Case II solution is unramified away from* ℓ*, the input the descent invokes a second time to seed its invariant.*

Theorem 0.2.3 (Situation from a rational solution). *A putative Case II solution, normalized to the real cyclotomic field, produces a "situation" of Washington 9.5 equipped with a measure* $m \geq p(p-1)/2$*.*

Theorem 0.2.4 (Principality from Vandiver (Washington 9.6)). *Under $p \nmid h^+$ alone, the ideal factors of the situation are principal — the step that classically needed full regularity, here powered by the certificate.*

Theorem 0.2.5 (One descent step). *From a situation the descent produces a new situation at a smaller parameter, via the mod-ℓ cancellations of Washington's Lemmas 9.6–9.7.*

Theorem 0.2.6 (The step decreases the measure). *The descended situation has strictly smaller measure — the well-foundedness that makes the descent terminate.*

Theorem 0.2.7 (No minimal situation). *No situation of minimal measure can exist; hence Case II admits no solution.*

0.3 Case I

Definition 0.3.1 (Case I hypothesis). `CaseIHolds` *p: no Case I solution ($p \nmid xyz$) exists. It is a hypothesis of the modular engine, discharged below by a certificate.*

Theorem 0.3.2 (B_{p-5} route to Case I). *If $p \nmid B_{p-5}$ then Case I holds — Kummer's 1857 refinement, used at the Wolstenholme primes where $p \mid B_{p-3}$ silences the primary criterion.*

Definition 0.3.3 (Sophie Germain certificate). `sgCert` *$p\,q$: the Boolean certificate that an auxiliary prime q meets Sophie Germain's conditions for p.*

Theorem 0.3.4 (Germain route to Case I). *If* `sgCert` *$p\,q$ holds for some auxiliary prime q, then Case I holds — the certified route at the two Wolstenholme primes.*

0.4 The crown theorems

Theorem 0.4.1 (Engine, modular form). *If* `SmallWitnessCert` *p holds and Case I holds at p, then* `FermatLastTheoremFor` *p: Case II is closed by the descent, Case I taken as hypothesis.*

Theorem 0.4.2 (Engine, discharged form). *Given the Q_i certificate, the size bound $\ell < p^2 - p$, and the irregularity certificate with $p-3 \notin L$,* `FermatLastTheoremFor` *p holds with nothing about Case I assumed — the non-Wolstenholme headline.*

Theorem 0.4.3 (Reduction to two open problems). *If every prime has a small-witness certificate and every odd prime has a Sophie Germain auxiliary, then Fermat's Last Theorem holds — with no modularity input. This is the paper's headline reduction; both hypotheses are classically open.*

Theorem 0.4.4 (Widened budget). *The discharged form with the budget $\ell < p^2 - p$ replaced by $2\ell \le 3p^2 - 5p + 2$ and $p \nmid (\ell-1)/p$, with no other change: the bad set of Washington's Lemma 9.7 is proved empty in that range (*`noBad_of_two_mul_le`*). Case I runs on a separate Sophie Germain prime.*

Theorem 0.4.5 (Bad-set certificate). *Case II with no size bound on ℓ: the Q_i certificate, $p \nmid (\ell-1)/p$, and the computable emptiness certificate* `noBadCert` *for the bad set of Lemma 9.7, together with a Sophie Germain prime for Case I, give* `FermatLastTheoremFor` *p.*

Theorem 0.4.6 (Joint auxiliary). *One prime ℓ carrying both certificates, the all-even Q_i test and the Sophie Germain conditions, with $p \nmid (\ell-1)/p$ and no upper bound on ℓ, gives* `FermatLastTheoremFor` *p: Legendre's condition (A) supplies the seed divisibility $\ell \mid z$ that the size bound used to buy.*